Paleomagnetism of the Late Precambrian Fourchu Group, Cape Breton Island, Nova Scotia

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Volcanogenic sediments of the Fourchu Group and a gabbroic intrusion, which are found in the Avalonian terrane of southeastern Cape Breton Island, have been sampled for paleomagnetic analysis. Upon detailed thermal and alternating-field demagnetization, three often-superimposed components of magnetization are obtained. One of these is aligned with the present-day geomagnetic field direction in Nova Scotia and is assumed to be of recent origin. The second group of directions is south-southeasterly and shallow, is postfolding in age, and is inferred to represent a Carboniferous overprint. The third direction, carried almost always by hematite, is also postfolding and yields a dual-polarity mean direction to the northwest or southeast, with a fairly steep inclination ($D = 132°$, $I = -63°$). This last direction is not seen in Avalonian or other North American rocks of Devonian or younger age; it is, therefore, bracketed in age between the earliest folding of the rocks and the latest Silurian. Given that Taconic folding has not been reported for this area, we assume that this magnetization was introduced in the rocks during uplift and oxidation after an Avalonian folding phase. For the Avalon terrane of Nova Scotia, the available paleomagnetic data reveal a set of moderately high paleolatitudes for the Late Precambrian and early Paleozoic, in contrast to the near-equatorial values predicted for the area under the assumption that it remained fixed with respect to the craton. On the other hand, strong similarities exist between Avalonian paleolatitudes and those for Armorica and Gondwana; a tentative reconstruction is proposed in which Avalon is adjacent to Armorica and Gondwana in the Late Precambrian and early Paleozoic.

Les sédiments volcanogéniques du groupe de Fourchu et une intrusion gabbroïque du terrane D'Avalon, dans le sud-est de l'île du Cap Breton, ont été échantillonnés pour l'analyse paléomagnétique. L'examen détaillé de la désaimantation thermique et par champ alternatif révèle trois composantes d'aimantation, souvent surimposées. Une de ces composantes est alignée avec la direction actuelle du champ géomagnétique en Nouvelle-Écosse, et dont l'origine semble récente. Le deuxième groupe de directions est sud-sud-est et de faible profondeur, d'âge postérieur à la déformation et considéré comme une surimpression datant du Carbonifère. La troisième direction, portée presque toujours par l'hématite, est également postérieure à la déformation, et elle fournit pour la polarité-double une direction moyenne nord-ouest ou sud-est et une inclinaison passablement abrupte ($D = 132°$, $I = -63°$). Cette dernière direction n'est pas observée dans les roches dévoniennes ou plus jeunes d'Avalon ou ailleurs en Amérique du Nord; par conséquent leur âge est confiné entre entre le début de la déformation des roches et la fin du Silurien. Étant donné que la déformation taconique semble absente dans cette région, nous supposons que cette aimantation a été introduite dans les roches durant le soulèvement et l'oxydation après la phase de plissement d'Avalon. Les données paléomagnétiques disponibles pour le terrane d'Avalon fournissent une série de paléolatitudes modérément élevées pour le Précambrien supérieur et le Paléozoïque inférieur, opposément aux valeurs attribuées à l'équateur pour la même région et qui sont fondées sur une immobilité des terrains d'Avalon par rapport au craton. D'autre part, il existe une ressemblance étroite entre les paléolatitudes d'Avalon et celles d'Armorica et de Gondwana; une reconstitution est proposée en juxtaposant les terrains d'Avalon à ceux d'Armorica et de Gondwana au Précambrien supérieur et au Paléozoïque inférieur.
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Introduction

The Avalon basement areas of easternmost Canada and eastern Massachusetts (Fig. 1*a*) show a marked contrast in geological facies, deformation episodes, ages, and faunal affinities when compared with the North American craton (Wilson 1966; McKerrow and Ziegler 1972; Whittington and Hughes 1972; Rast *et al.* 1976; Keppie 1977; Williams 1978, 1984; Rao *et al.* 1981; Cocks and Fortey 1982). It is, therefore, not surprising that the Avalonian has been recognized as a suspect and probably displaced terrane (Zen and Palmer 1981; Williams and Hatcher 1983; Neuman 1984; Williams 1984).

There are, however, two unresolved questions regarding the history of this area, and paleomagnetic techniques appear to be suitable for eventually providing answers to both. The first question concerns the early Paleozoic position of the Avalonian terranes. Given that it was not part of North America, did Avalon constitute an independent microplate, or was it attached to another major cratonic block? Well-dated paleomagnetic results can record the displacements of tectonic elements such as Avalon, through apparent polar wander paths (APWP's) and thus constrain models for the history and original affinity of the terrane. The second question regarding the history of Avalon concerns the timing of its collision with the North American craton; assuming that this collision can be recognized in orogenic phases in the area, the Taconic, Acadian, and Alleghenian orogenies are all possible candidates. In this case, paleolatitude determinations from paleomagnetic data can quantify minimum displacements between terranes to help identify the earliest time at which a collision was possible.

To contribute to our knowledge regarding these questions, a paleomagnetic sampling program was initiated, and Late Precambrian, Cambrian, Late Ordovician, Devonian, and Early Carboniferous rocks were collected. We report here the results for the Late Precambrian Fourchu Group of eastern Cape Breton Island; our results for the other formations have been reported elsewhere (Johnson and Van der Voo 1983, 1985; Scotese *et al.* 1984; Van der Voo and Johnson 1985). In addition, apparently unremagnetized paleomagnetic results are available for Paleozoic intrusive and sedimentary rocks from the Avalon Zone (Rao *et al.* 1981; Kent 1982; Irving and

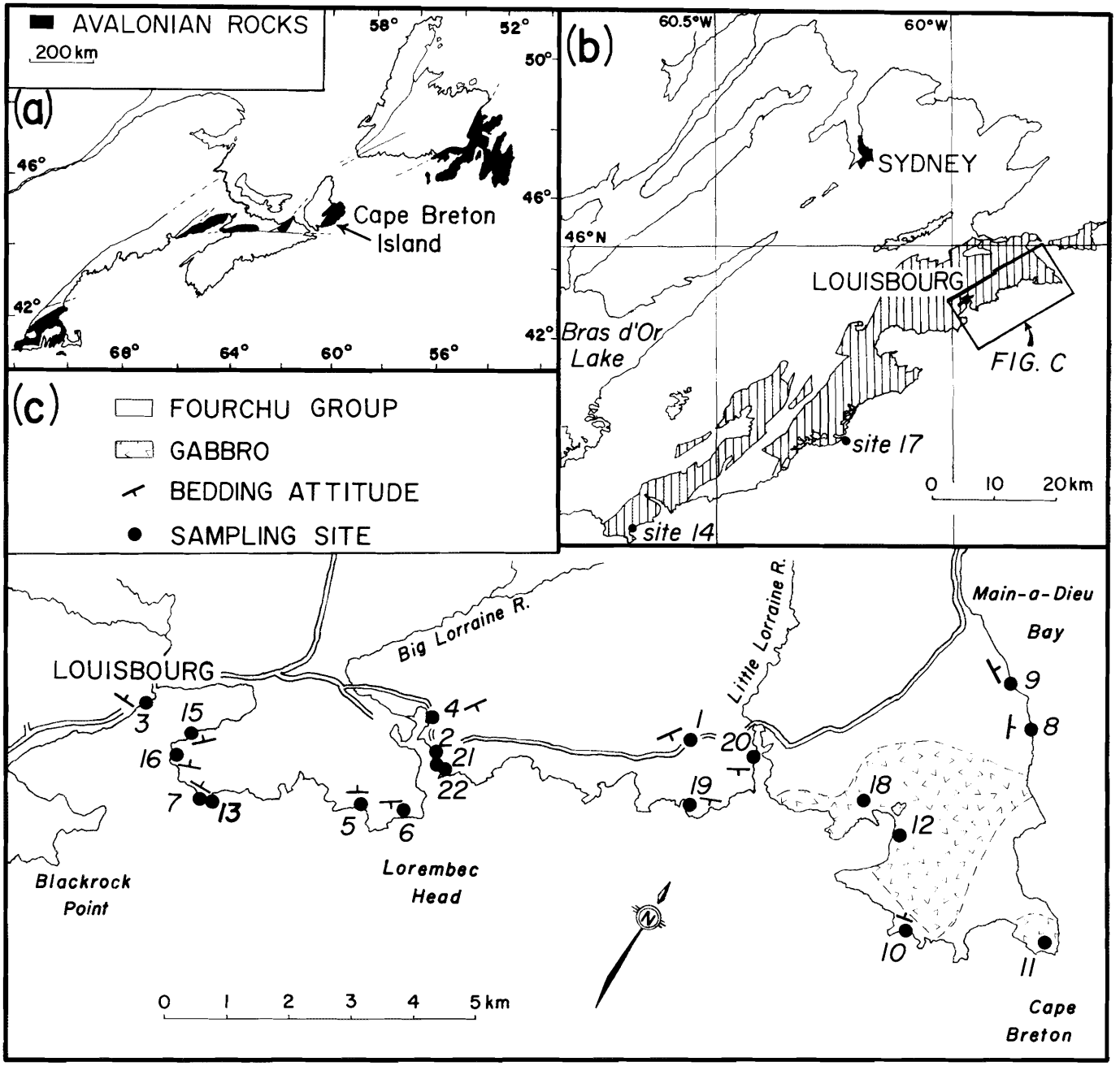


FIG. 1. Regional tectonic setting of Avalonian rocks in the northern Appalachians (*a*), and local geologic maps of the Fourchu Group in southeastern Cape Breton Island (*b*, *c*) showing paleomagnetic sampling sites (after Keppie 1979; Murphy 1977).

Strong 1985) and for several pre-Carboniferous formations from adjacent zones (Deutsch and Rao 1977; Lapointe 1979; Roy *et al.* 1979; Roy and Anderson 1981; Kent and Opdyke 1980). Collectively, these data are building a base of well-dated and reliable paleomagnetic results that constrain the multitude of hypotheses and models for the plate tectonic Paleozoic history of the northern Appalachians.

Geological setting and sampling

The Fourchu Group outcrops primarily in southeastern Cape Breton Island (Fig. 1), where it comprises interbedded tuffs, agglomerates, minor volcanic flows, and volcaniclastic sediments (Weeks 1954; Murphy 1977). Felsic tuffs and coarse agglomerates are the dominant rock types in the Fourchu Group, but minor mafic tuffs or flows are present throughout the outcrop area. Individual units, which are recognized on the basis of uniform composition or texture, range in thickness from 1 to ~60 m, although their thickness often varies laterally. Whereas all of the units in the southern part of the outcrop area are inferred to have been deposited subaerially, well-developed bedding and cross-stratification in tuffs in the northern part of the area indicate shallow-water deposition. Lithologies in the waterlaid sediments are much better sorted, and units are laterally continuous, allowing an internal stratigraphy to be recognized (Murphy 1977).

Although not exposed in contact with other rock units, the Fourchu Group is thought to underlie a sequence of Eocambrian and fossiliferous Lower Cambrian strata that also occur in this part of Cape Breton Island. This is based on inference

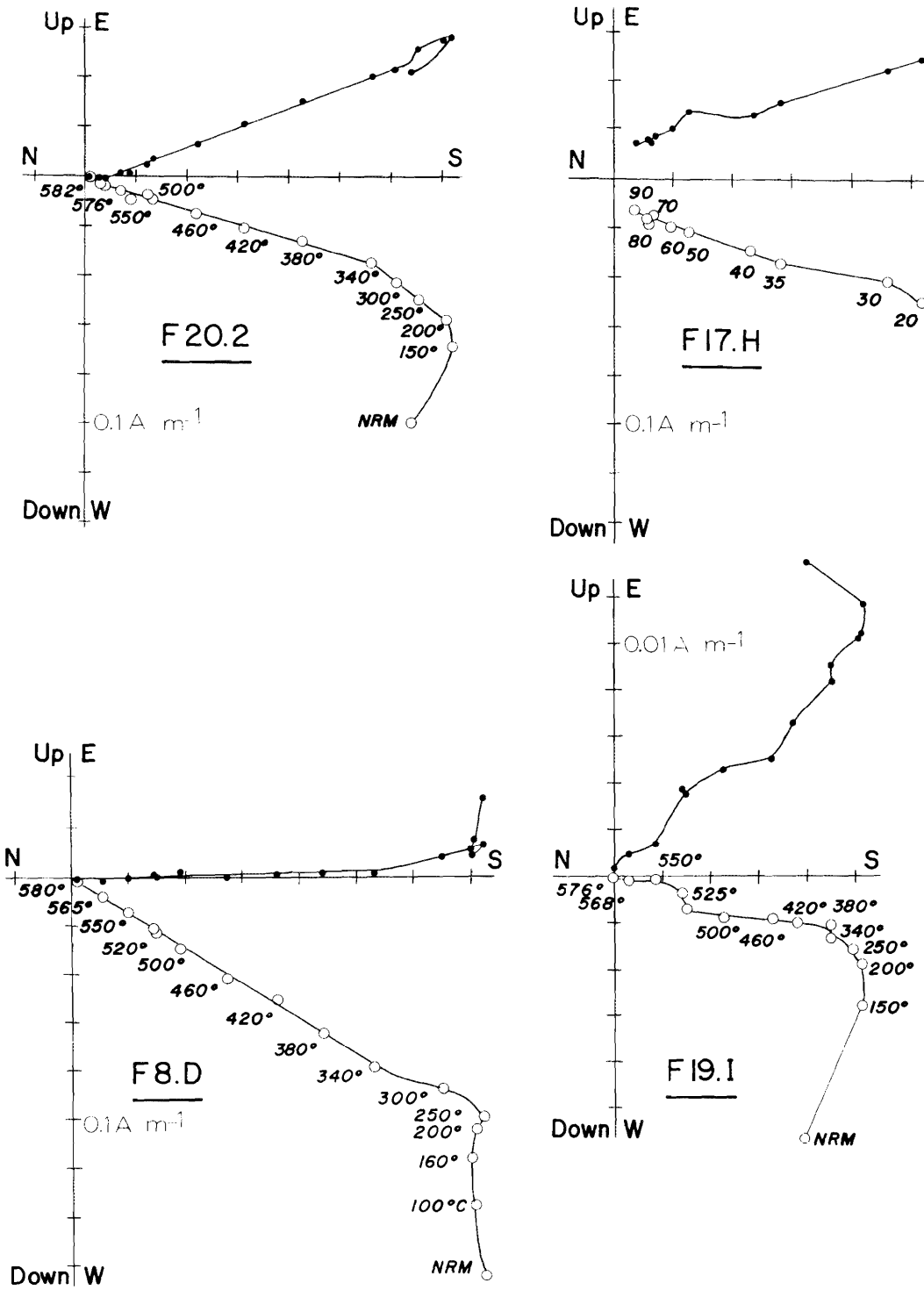


FIG. 2. Orthogonal vector diagrams (Zijderveld 1967) of stepwise thermal and alternating field (AF) demagnetizations of samples in which only the B component of magnetization is found. Open (closed) symbols represent projections onto the north–south vertical (horizontal) plane; the treatment applied to samples in successive steps is indicated in each diagram. The numbers to the left of the period in the sample name correspond to the number of the sampling site. In this figure and in successive figures with Zijderveld diagrams, sample directions are shown uncorrected for structural tilt.

from the geologic structure at several localities where Eocambrian strata are exposed nearby the Fourchu, as well as on the occurrence of rounded pebbles and cobbles of deformed Fourchu volcanics in conglomerate units within the Eocambrian – Lower Cambrian sequence (Smith 1978). The Fourchu has been correlated with lithologically similar volcanic strata occurring in eastern Newfoundland and in New Brunswick, which also appear to occupy the same stratigraphic position as the Fourchu (Weeks 1954; Rast *et al.* 1976; O'Brien *et al.* 1983). The volcanics in Newfoundland and New

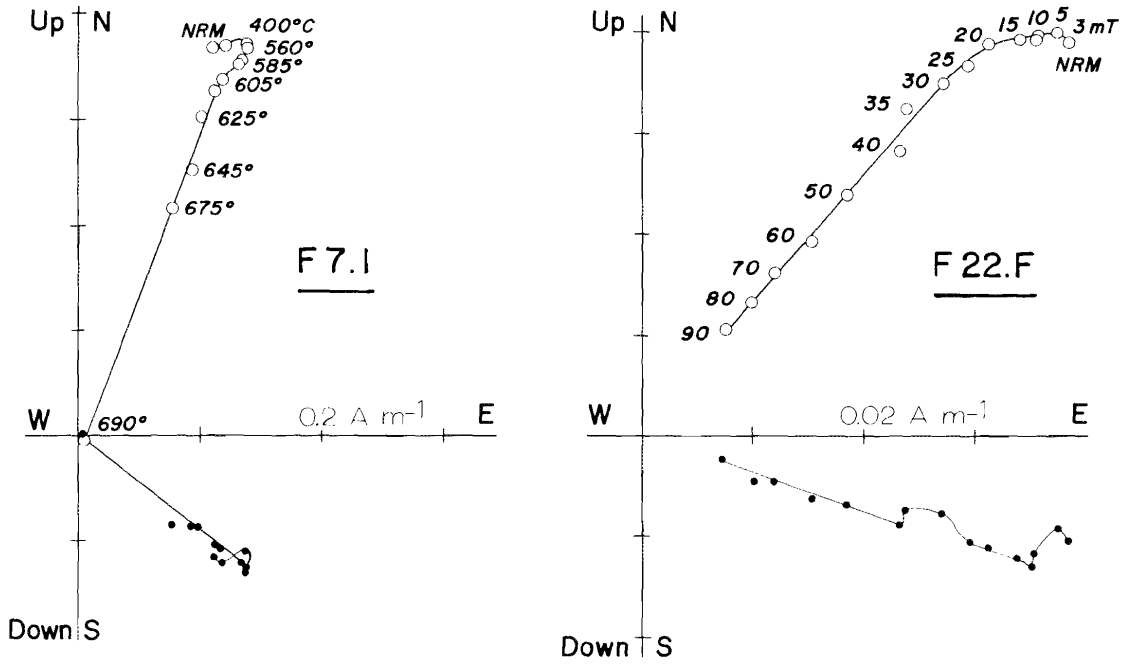


FIG. 3. Orthogonal vector diagrams showing thermal and AF demagnetizations of Fourchu samples that have only the A component of magnetization. In this figure, the magnetization vectors are projected onto the east–west vertical (open symbols) and horizontal (closed symbols) planes to improve clarity.

Brunswick have yielded radiometric ages in the range 600–630 Ma, and a similar age is considered likely for the Fourchu Group (O'Brien *et al.* 1983).

Deformation into relatively open folds is apparent for the well-stratified units of the Fourchu in the northern part of the study area. Penetrative deformation (axial-planar schistosity) is developed only locally in these strata, but it is commonly developed in more massive units throughout the sampling area. Based on the occurrence of cobbles of deformed Fourchu volcanics in Eocambrian strata in southeastern Cape Breton Island (Smith 1978), deformation is taken to be Precambrian and may be correlated with similar Late Precambrian disturbances recognized in other parts of the Avalon terrane (Weeks 1954; Strong 1979). The well-rounded pebbles and cobbles of the Fourchu in the Eocambrian strata indicate exposure of the Fourchu source area following deformation, and some authors have suggested that the conglomerates and associated red sandstones and siltstones represent postorogenic molasse deposits (Rast *et al.* 1976). A gabbroic body in the northern part of the study area (Fig. 1*c*) has been shown to intrude the youngest volcanogenic sediments in this region, but deformation fabrics in comagmatic dikes similar to those found in the surrounding Fourchu country rock suggest a Precambrian age for these intrusives (Murphy 1977). We note that later Acadian age deformation in this region is locally characterized by simple tilting of Paleozoic strata between basement blocks that do not seem to have suffered internal deformation themselves (Weeks 1954).

Paleomagnetic sampling was concentrated in the tuffaceous sediments in the northern part of the outcrop area (Fig. 1*c*), where the structural attitudes of the strata are readily measured and a fold test is possible. In addition, subaerial volcanics were sampled at sites 7, 14, and 17. Reddish hematite staining is present in virtually all of these strata to a greater or lesser degree, but the timing of hematite formation could not be assessed from field observations. Fourchu volcanics also occur inland but in isolated settings that are not ideal, and therefore they have not been sampled. The gabbro was sampled at Baliene Cove (sites 12 and 18) and at Cape Breton (site 11). Oriented samples were taken with a portable gasoline-powered drill where possible, but in places that were inaccessible with the drill, block samples were collected. All samples were oriented with a magnetic compass and inclinometer; the intensity of the natural remanence is too low to influence the compass readings.

Paleomagnetic results

The intensities of the natural remanent magnetizations (NRM) range between 10^{-4} and 10 A/m, as measured on either a Schonstedt spinner magnetometer or a ScT cryogenic magnetometer. Thermal and alternating field (AF) demagnetizations, carried out on a pilot collection of samples with Schonstedt equipment, proved to be of equal value and yielded very similar results; nevertheless, most samples were treated thermally, because complete decay of the NRM could be achieved.

Demagnetization results were plotted in orthogonal vector diagrams (Zijderveld 1967), and linear trajectories on these diagrams were used to determine directions of magnetic components, through vector subtraction comprising, generally, four measurements or more. The demagnetization diagrams typically revealed up to three vectorial components of magnetization. Similar components were observed in samples of widely differing intensities and in the gabbro as well as the tuffs. A softer, low blocking temperature or low-coercivity component was first removed, typically by 340°C or 20 mT (Fig. 2). Its direction in *in situ* coordinates generally conforms to the local present-day field direction, and it is likely to be a viscous remanence of recent origin. The second and third

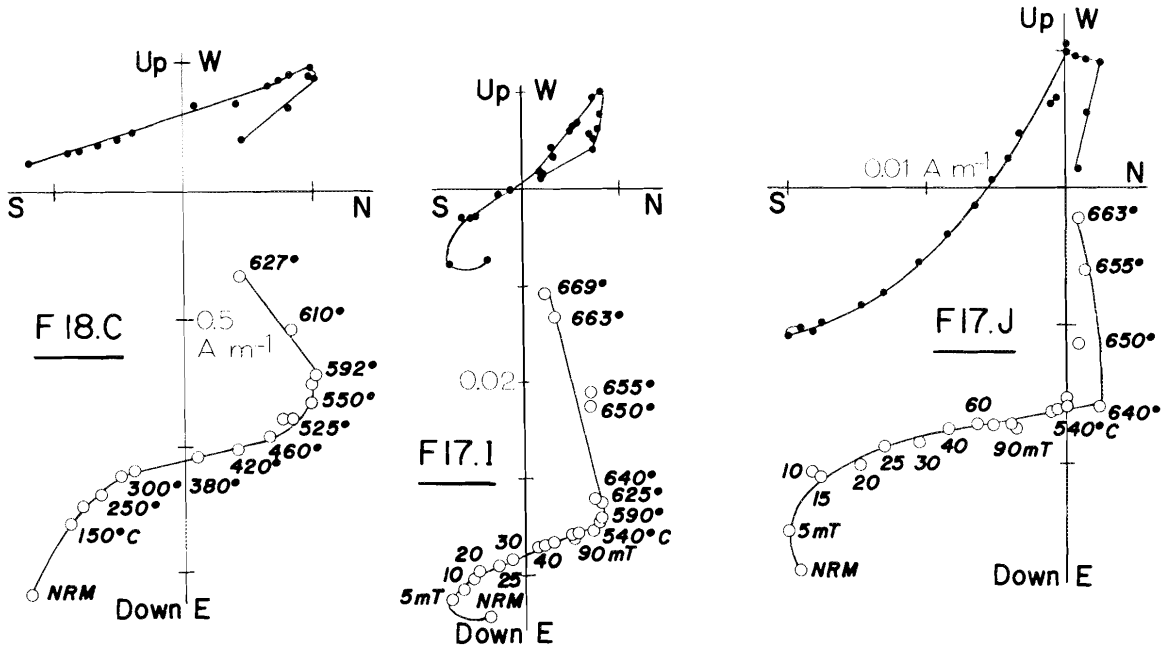


FIG. 4. Orthogonal vector diagrams of thermal and combined AF–thermal demagnetizations of samples in which both A and B components are resolved. The plotting conventions used are the same as in Fig. 2, i.e., open (closed) symbols represent projections onto the north–south vertical (horizontal) plane.

directions sometimes occur together, sometimes alone in a given specimen. We will first discuss the cases in which only one of these two components has been obtained, and then describe the results from the composite NRM's.

A little less than half (59) of the total number of samples demagnetized (135) reveal only a single, southerly component in addition to the viscous remanence, and this component (*in situ*) has a shallow, mostly downward, inclination. We call this the B component, and Zijderveld plots showing this component are presented in Fig. 2. Generally, this component decays linearly to the origin, although on some occasions there are hints of another component remaining in the sample after the last treatment (e.g., sample F17.H in Fig. 2). The blocking temperatures and AF decay of this component are typical (but not diagnostic) of magnetite. As we will see below, there are supporting arguments for magnetite as the carrier of this magnetization.

At least 14 samples from four sites reveal only the third characteristic direction, as illustrated in Fig. 3. These directions (*in situ*) are east-southeasterly, with fairly steeply upward inclination (Fig. 3), or antipodal to this, with west-northwesterly declinations and steeply downward inclinations. We call this the A component, and it always decays linearly to the origin. The blocking temperatures of this component are typical of hematite (560–680°C).

A large number of samples, however, reveal both A and B components superimposed in the same specimen. Zijderveld diagrams for such samples are presented in Fig. 4. It can be seen in these diagrams that the blocking temperatures for these components agree with those discussed above for the cases of univectorial behavior and separate into ranges typical for magnetite and hematite, respectively.

As is so often the case with multivectorial NRM's, perfect isolation of the two individual components is not always achieved. This is the reason, primarily, for discussing first the nearly univectorial samples, to establish the "credentials" of the A and B components. Where the coercivities or blocking temperatures of these two components apparently overlapped, an intermediate-direction trajectory could be recognized in the Zijderveld plots, and often these trajectories were curved. Examples of such behavior, where neither the A nor the B component, apparently, was isolated, are illustrated in Fig. 5. It can be seen in this figure that perfectly linear segments are not observed in the demagnetizations of these samples, and consequently no directions could be isolated from them. In Table 1, which presents a compilation of all A and B site-mean directions, these samples are not included (the rejected samples are reflected in the column reporting the ratio of samples used for the computation of the mean to the total number of samples analyzed).

Figures 6 and 7 present equal-area plots of the B and A sample directions, respectively, that were used for the computation of a mean; in addition, site means (as listed in Table 1) are shown in *in situ* and bedding-corrected coordinates. It can be seen at a glance that both the B and the A site-mean directions disperse greatly upon the application of the bedding correction, and we conclude that both these characteristic directions are postfolding in age. The failure of the fold test is statistically significant in both cases, according to the criteria of McElhinny (1964), as determined from the ratios of precision parameters for site means before and after bedding correction given in Table 2.

To conclude this section, we wish to present some additional evidence for our identification of the carriers of magnetization from experiments on acquisition of isothermal remanent magnetization (IRM) (Fig. 8.). Representative samples from a majority of sites were selected for IRM acquisition in fields up to 3.5 T (35 000 Gs). Most of these samples showed saturation in high fields only, indicative of coercivities well beyond the theoretical maximum for magnetite. Given the high blocking temperatures (up to 680°C), clearly hematite is a major carrier of the NRM in these rocks. However, a few samples (e.g.,

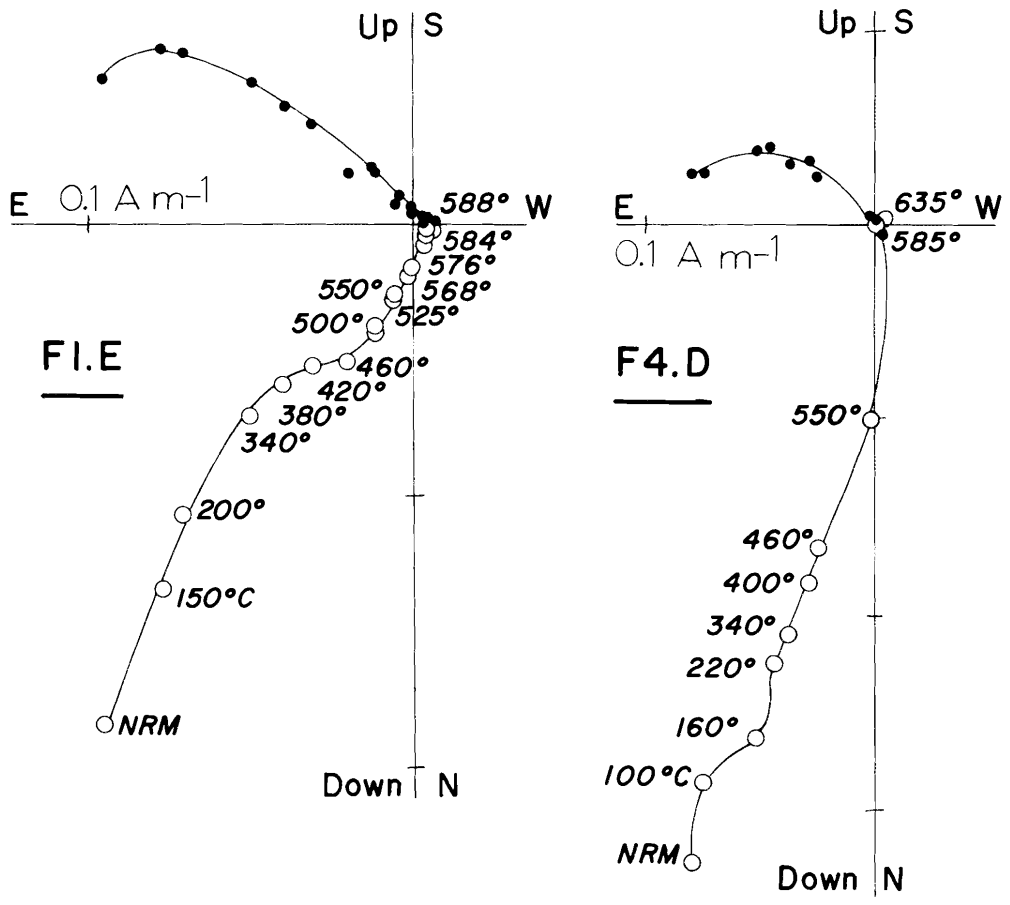


FIG. 5. Orthogonal vector diagrams of thermal demagnetizations of samples in which no magnetization components are resolved. The plotting conventions used are the same as in Fig. 3.

sample F5.1 in Fig. 8*a*) showed rapid saturation (below 0.3 T), blocking temperatures below 575°C, and AF demagnetization decay down to the origin below 100 mT, indicating that only magnetite is present. In addition, many samples showed IRM acquisition like that illustrated in Fig. 8*b*, i.e., after an initial and fairly rapid acquisition in fields below 0.3 T, the slope of the acquisition of the curves begins to flatten, yet the samples do not saturate until 2 T. Back fields were applied to these samples to measure their coercivities, H_{cr}; the resultant back-field IRM curves given in Fig. 8*b* indicate the presence of two carriers with coercivities (H_{cr}) of less than 50 mT and greater than 0.3 T, respectively. We interpret this behavior as the effect of coexisting magnetite and hematite in these samples. Given that both the A and the B components are postfolding, both NRM carriers can be safely assumed to be either fully secondary or thermally remagnetized.

Interpretation of results

The mean directions of the A and B components of magnetization, their associated statistical parameters, and the calculated pole positions are given in Table 2. The B direction and its paleopole have been recognized (and often as overprints) in many previous studies of rocks in the Appalachians; we conclude, in agreement with the conclusions of many other studies (e.g., Johnson and Van der Voo 1983; Roy and Morris 1983; Irving and Strong 1985; Spariosu *et al.* 1984; Van der Voo and McCabe 1985), that the B magnetization represents a Carboniferous overprint. We note that there are a few normal-polarity directions in the B population (Fig. 6), but these cannot really be used to constrain the age of the B magnetization because normal polarities are known to occur before, as well as during, the Late Carboniferous – Early Permian Kiaman interval. On the other hand, the paleolatitude derived from the paleopole location (34°N, 139°E) is about 9°S, and this would suggest an age of magnetization coinciding with the Late Carboniferous Westphalian C–D substages, according to the compilation of Roy and Morris (1983). One may well pose the question, what process gave rise to this overprint? Assuming that we are correct in our identification of magnetite as the carrier, two options are available: a thermal partial remagnetization or a chemical remanence resulting from secondary growth of magnetite. A priori, a thermal overprint seems more likely, as a side effect of the Alleghenian–Hercynian orogeny, but theoretical considerations (Pullaiah *et al.* 1975) would require rather high temperatures to completely reset the magnetite (no A magnetization is observed with blocking temperatures less than 560°C).

The A direction, on the other hand, is not so easily dated. This fairly steep direction is not seen or expected in Avalonian rocks or in rocks from adjacent Appalachian zones or from the North American craton, for any period after the Silurian. The A magnetization postdates the Fourchu Group's folding, which itself is not well constrained in age but is not likely to be Taconic (Smith 1978; Williams and Hatcher 1983) and cannot be Acadian if our assumption of a pre-Devonian age of the A magnetization is correct. We infer that the principal deformation of the Fourchu Group has been the result of the Avalonian

TABLE 1. Summary of site-mean paleomagnetic data

		A direction				B direction			
Site	$N_A/N_B/N_0$	D/I (°) (before TC)	D/I (°) (after TC)	k	α_{95} (°)	D/I (°) (before TC)	D/I (°) (after TC)	k	α_{95} (°)
1	9/7/13	316/+60	349/+83	32	9	154/+24	152/+3	85	7
2	0/4/ 9	—	—	—	—	172/+24	161/+10	29	17
3	2/3/ 7	124/−60	141/−12	—	—	179/+17	228/+66	36	21
4	0/2/ 4	—	—	—	—	154/+14	153/−8	—	—
5	0/2/ 2	—	—	—	—	182/+13	260/+54	—	—
6	1/2/ 3	146/−72	31/−77	—	—	174/−2	174/−28	—	—
7	3/0/ 3	135/−62	161/−23	6566	2	—	—	—	—
8	0/3/ 3	—	—	—	—	183/+31	167/+32	331	7
9	0/4/ 4	—	—	—	—	170/+24	154/+44	154	7
10	5/1/12	309/+58	324/+34	22	16	163/+39	158/+66	—	—
11	3/0/ 4	140/−61	140/−61	156	10	—	—	—	—
12	0/5/ 6	—	—	—	—	156/+14	156/+14	25	16
13	4/0/ 4	156/−67	160/−41	34	16	—	—	—	—
14	0/6/ 6	—	—	—	—	173/+10	175/+17	19	16
15	1/7/ 9	333/+59	334/+12	—	—	353/+2	1/−37	11	21
16	0/3/ 3	—	—	—	—	157/+9	148/+57	54	17
17	3/3/ 6	284/+65	160/+32	77	14	143/+18	143/−58	71	15
18	1/4/ 8	326/+52	326/+52	—	—	153/+1	153/+1	141	8
19	2/7/ 9	302/+72	205/+62	—	—	151/+12	151/−17	40	10
20	0/8/11	—	—	—	—	160/+23	153/+13	51	8
21	5/0/ 5	293/+58	315/+58	58	10	—	—	—	—
22	4/0/ 4	103/−66	76/−78	31	17	—	—	—	—

NOTES: $N_A/N_B/N_0$ is the ratio of samples in which a direction could be isolated (either A, B, or both) to the number of samples demagnetized; D and I are declination and inclination of the paleomagnetic direction, before or after correction for the tilt of the strata (TC); k and α_{95} are statistical parameters associated with the means (Fisher 1953).

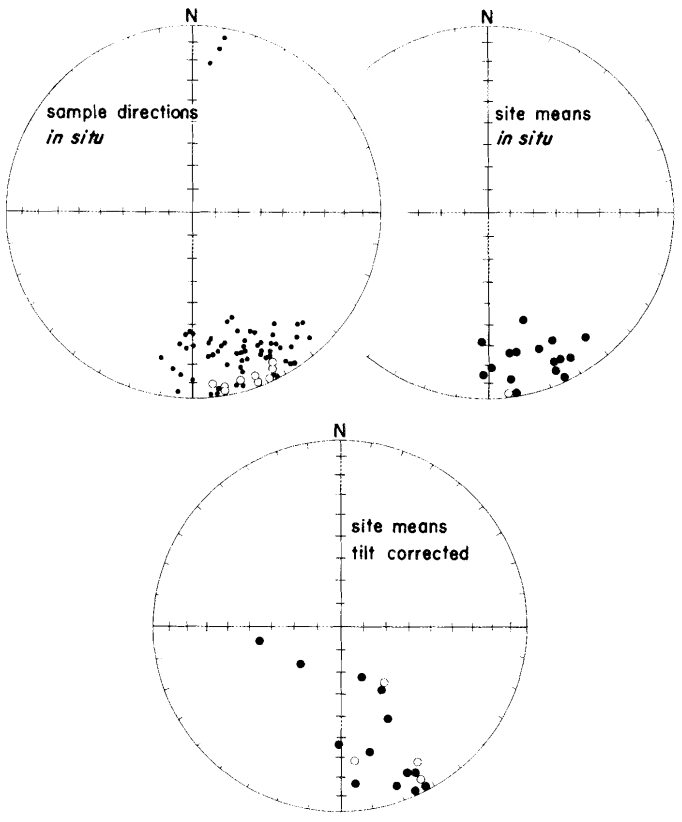


FIG. 6. Equal-area projections of the B component sample directions (*in situ* only) and site-mean directions for the B component, both *in situ* and tilt corrected. Closed (open) symbols represent projections on the lower (upper) hemisphere.

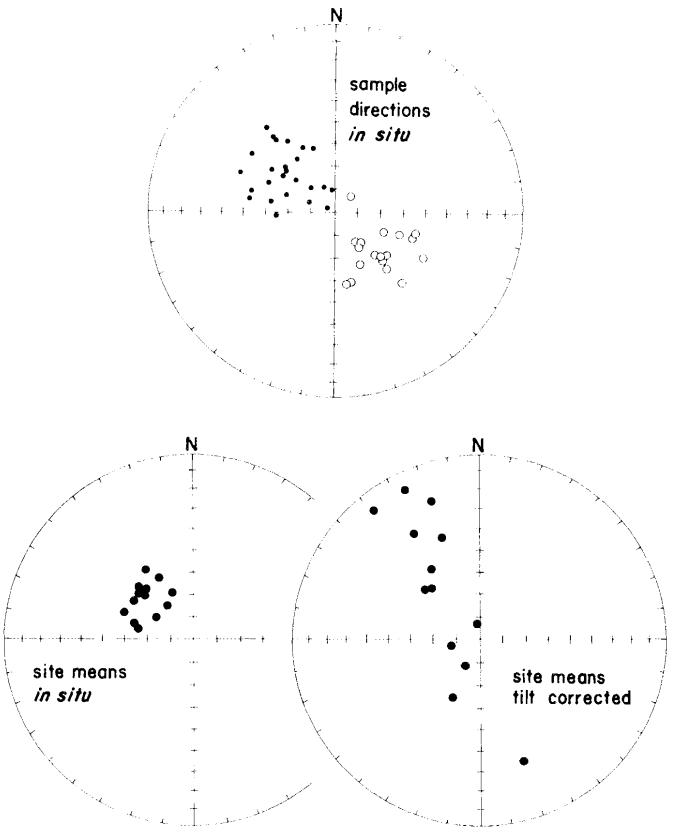


FIG. 7. Equal-area projections of *in situ* sample directions for the A component and A component site means, both *in situ* and tilt corrected.

TABLE 2. Summary of mean directions and poles

Component	N/N_0	D/I (°) (before TC)	α_{95} (°)	k_1	k_2	k_1/k_2	Paleopole (°)
A	13/22	132/−63	5.0	70.5	4.3	16.3	56N, 225E
B	17/22	164/+16	7.2	25.7	3.8	6.7	34N, 139E

NOTES: N/N_0 is the ratio of site-mean directions used to compute the mean to the total number of sites; D and I are the declination and inclination of the mean paleomagnetic direction, before correction for the tilt of the strata (TC); α_{95} and k are statistical parameters (Fisher 1953), with k_1 (k_2) the precision parameter before (after) correction for the tilt of the strata.

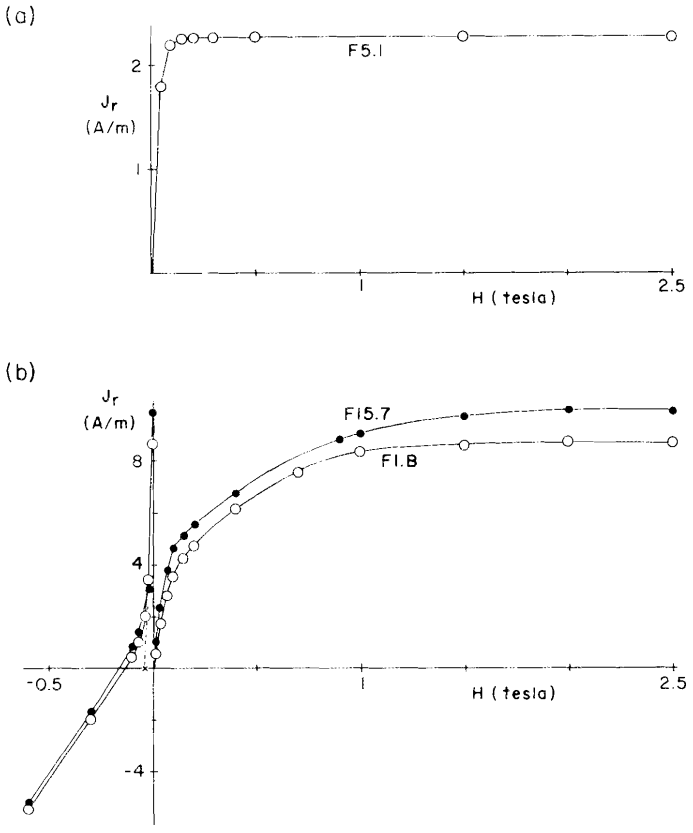


FIG. 8. Isothermal remanent magnetization (IRM) acquisition curves for (*a*) a sample that achieves saturation magnetization in applied fields of less than 0.3 T and (*b*) samples that display rapid initial acquisition of IRM but do not saturate until 2 T. H is the intensity of the magnetic field applied to samples (at 20°C), in either a positive or negative (back-field) direction; J_r is the remanence intensity with corresponding positive or negative direction. In (*b*), the initial slopes of the back-field curves for samples are projected to the horizontal axis (dashed line) to estimate the remanent coercive force H_{cr} for the low-coercivity component in these samples.

(latest Precambrian) orogeny. Thus, the A magnetization is only bracketed in age between the latest Precambrian and the Late Silurian. A more precise dating of this magnetization is impossible, but tentatively we conclude that an Early Cambrian age may be preferred. Our rationale for this preference is that the A magnetization, carried by hematite in these volcanogenic sediments, is undoubtedly the product of a secondary oxidation process; conditions for such oxidation are typically most favorable during erosion and uplift following the folding event. We note that this is consistent with local geologic evidence for exposure of the Fourchu after deposition but prior to the deposition of Eocambrian–Cambrian conglomerates and associated red sediments containing detritus derived from the Fourchu.

Avalon paleolatitudes

The Fourchu A component direction of $D/I = 312°/+63°$ corresponds to a F1 paleopole at 56°N, 225°E and a paleolatitude of 44.5° (Table 2). Given the inferred latest Precambrian to early Paleozoic age of this pole and the uncertain tectonic history of the Avalon continent, the polarity of the F1 pole is not immediately evident, and it may thus represent either the north (normal polarity) or the south (reverse polarity) paleomagnetic pole with respect to Nova Scotia. The polarity assignment is clearly important for the tectonic history of Avalon because in the former case Nova Scotia would have occupied a paleolatitude position of 45°N, whereas in the latter case it was located at 45°S. Although the polarity for the F1 pole cannot be determined independently, some constraint on its correct assignment is provided by comparison with other well-dated magnetizations in lower Paleozoic strata from Nova Scotia. The Mid-Cambrian Bourinot Group (Johnson and Van der Voo 1985) and the Upper Ordovician Dunn Point Formation (Van der Voo and Johnson 1985) yield directions with inclinations similar to that of the Fourchu A component and, therefore, indicate similar paleolatitudes for Nova Scotia, even though their declinations differ by up to 160° from the Fourchu result. The steeply downward directions in the Paleozoic rocks have previously been interpreted as reverse polarity magnetizations, thus placing Nova Scotia in intermediate southern paleolatitudes in the early and early mid-Paleozoic. This choice of polarity for the Bourinot and Dunn Point results minimizes the Paleozoic latitudinal drift of Nova Scotia as well as the latitudinal separation between Nova Scotia and North America. If this assignment of polarity is correct, then a similar polarity for the Fourchu A magnetization is the only reasonable choice, given that the age difference between the F1 pole and either the Bourinot or the Dunn Point pole is at most 70 Ma: the alternative would imply highly unlikely drift rates of some 10 000 km per 70 Ma, i.e., at least 14 cm/year. We thus infer a position for Nova Scotia at 45°S paleolatitude at the time of the acquisition of the Fourchu magnetization. Given that a large relative rotation between the Bourinot strata in central Cape Breton Island and the Fourchu Group rocks can be ruled out (e.g., Hutchinson 1952), the difference in paleomagnetic declinations between these rock units can only be interpreted as apparent polar wander with respect to Nova Scotia, presumably in the Early Cambrian. This may have involved appreciable drift of Nova Scotia (Avalon), depending on the position of the tectonic rotation (Euler) pole with respect to Nova Scotia, but it certainly could be accommodated by a much smaller amount of drift than the 10 000 km that would

result from postulating the opposite polarity for the Fourchu A component, as discussed above.

The available Late Precambrian and early Paleozoic paleomagnetic data from the Avalon terrane in Nova Scotia and Newfoundland are listed in Table 3. In addition to those results already discussed, this table includes data for three different magnetization components observed in several granites from Cape Breton Island (Rao *et al.* 1981) and for three rock units of the Marystown Group, a unit correlative with the Fourchu in eastern Newfoundland (Irving and Strong 1985). The result for the Ordovician Wabana – Bell Island Formation is not included in Table 3 because of the likely remagnetization of this rock unit (e.g., Buchan and Hodych 1982). Considering the granite magnetizations first, we note that they are rather poorly dated, as are the granites themselves; the ages given in the table for individual magnetizations correspond to radiometric ages of the plutons in which they are found, including the associated error (Cormier 1972). The CB component occurs only in an older suite of granites where it is a common component, yielding a rather shallow direction and thus indicating a low paleolatitude for Nova Scotia. The possible age range for the CB component is particularly broad, and although it partly overlaps with ages for the Fourchu and Bourinot magnetizations, it may, in fact, be much older than both of these. A second component, CB1, is reported as a single component from a younger suite of granites in northwest Cape Breton, apparently constraining it to be at least as young as the date for these granites (410 ± 50 Ma), although it is also documented in a number of the older plutons as an overprint component. It has a direction with a steep inclination, consistent with those found for early Paleozoic strata whose age it partly overlaps. The third granite component, CB2, is documented only from the Cape Smokey pluton (Rao *et al.* 1981, Fig. 4), where it is reportedly found at three sites. The Cape Smokey pluton is a foliated granite in northwest Cape Breton giving a somewhat anomalous (whole-rock isochron) age of 460 ± 60 Ma. Although this date may represent the age of emplacement of the granite, according to Cormier (1972) this age probably reflects partial resetting of the isotopic system in an older granite, as supported, in particular, by the occurrence of consistently young ("Acadian") biotite mineral ages in the granite and nearby older gneisses. In the case of a reset age for the granite, the magnetization may or may not have been reset. Although this leaves open the possibility of an older age for the CB2 component, it is tentatively assigned an age corresponding to the radiometric date obtained for the granite itself. The CB2 component in the Cape Smokey granite has a steep inclination observed with both polarities, although its declination is again different from that of the CB1 magnetization, as well as the Bourinot and Dunn Point directions. Finally, we consider the paleomagnetic results for the Marystown Group from eastern Newfoundland. All three rock units studied by Irving and Strong (1985) yield directions with similar intermediate inclinations but fairly widely dispersed declinations after tilt correction. These authors suggested, nevertheless, that field evidence indicates primary, or at least early, acquisition of magnetization in hematite in these rocks. Given that the sampling localities in the Burin Peninsula are widely separated, the declination dispersal may simply be a result of local tectonic rotations. However, because paleomagnetic inclinations are unaffected by rotations about vertical axes, the corresponding intermediate paleolatitudes can be thought of as representative for eastern Newfoundland for the time interval 620–600 Ma. Paleolatitude data are similarly unaffected by possible relative rotations between Avalon terranes in Nova Scotia and Newfoundland, and the results for the Marystown Group should, therefore, indicate the correct paleolatitude of the entire Avalon continent. Reassuringly, the paleolatitude for Avalon, according to the Marystown result, is not very different from that suggested by the Fourchu A component.

In summary, the Late Precambrian and early Paleozoic results from Nova Scotia and Newfoundland indicate fairly uniform intermediate southern paleolatitudes for Avalon throughout the interval ~600–400 Ma, with the CB component as the only exception. If an older age for the CB pole is correct, however, then Avalon may have drifted from lower to intermediate paleolatitudes sometime between 650 and 600 Ma.

Tectonic implications of Avalonian paleolatitudes

Methods of comparing paleomagnetic data from two or more blocks, when needed to test geotectonic models, have commonly involved the use of paleopoles and APWP's in the past. For the Avalonian basement areas, however, the relatively small number of available poles, as well as the real possibility of some relative rotations on a local scale, makes the APWP approach undesirable. Instead, it appears to be advisable and a priori more rigorous to restrict the comparison to paleolatitudes only; this not only avoids the uncertainties associated with rotations but also eliminates the need to place the paleopoles in the strict temporal sequence necessary to construct an APWP through them. We believe the paleolatitude-only approach to be more rigorous because it allows us to use paleolatitude in combination with pole age (time) in a single frame for a given block and to make proper use of the uncertainties associated with both these variables, something that is not possible in the APWP approach. Although the exercise may be more rigorous, however, we note that the similarities inferred from a paleolatitude comparison must, by necessity, be less compelling (permissive rather than diagnostic) than in the case of APWP comparisons. Thus, it appears obvious that the latter approach remains our eventual goal for future research.

For this study, we will compare the Avalonian paleolatitudes of Table 3 with the latitude positions occupied by the major Atlantic-bordering continents in the Late Precambrian and early Paleozoic. It may then be possible to infer which of these continents, if any, could have been together with Avalon during this time, based on their similar paleolatitudes. The possibility of similar paleolatitudes for Avalon and other Atlantic-bordering continents is tested by calculating predicted paleolatitudes corresponding to an expected position of Avalon relative to each continent and comparing these with observed data for Avalon. If the observed and predicted paleolatitudes are not different from one another, then we can conclude that the chosen fit is a permissible reconstruction.

Only three continental areas in the Atlantic domain have a reasonable abundance of reliable data for the period of interest: North America, Gondwana, and the Armorican Massif in France, even though the age distribution of the data from each of these regions differs slightly. We have chosen the fit of Bullard *et al.* (1965) as the basis for our calculation of predicted paleolatitudes, motivated in part by ease of computation but also guided by geologic similarities, particularly between Avalon, Armorica, and north Africa (e.g., Rast *et al.* 1976). The fit, moreover, resembles Late Precambrian reconstruc-

TABLE 3. Late Precambrian – Early Paleozoic paleolatitudes for Avalon

Formation	Symbol	Age (Ma)	D/I (°)	Paleopole (°)	Φ (°)	dp (°)	Reference
Nova Scotia							
Cape Breton granites	CB	650–490	318/+15	37N, 176E	8	2	Rao *et al.* (1981)
Cape Breton granites	CB1	410±50	145/+71	16N, 319E	55	6	Rao *et al.* (1981)
Cape Breton granites	CB2	460±60	256/+73	32N, 263E	59	7	Rao *et al.* (1981)
Fourchu	F1	610–540	312/+63	56N, 225E	45	6	This study
Bourinot volcanics	BV	540	113/+66	21N, 340E	49	11	Johnson and Van der Voo (1985)
Bourinot sediments	BS	540	87/+65	33N, 354E	47	11	Johnson and Van der Voo (1985)
Dunn Point	DP	437±15	155/+61	2N, 316E	42	5	Van der Voo and Johnson (1985)
Newfoundland							
Marystown, Calmer*	—	625–600	298/+54	32N, 231E	36	6	Irving and Strong (1985)
Marystown, Famine Back Cove*	—	625–600	133/+53	1N, 342E	34	5	Irving and Strong (1985)
Marystown, Garnish*	—	625–600	151/+53	5N, 329E	34	7	Irving and Strong (1985)

NOTES: D and I are declination and inclination of the mean paleomagnetic direction; Φ is paleolatitude of the sampling site; dp is the error in paleolatitude (McElhinny 1973) corresponding to a semi-axis of the oval of 95% confidence about the pole. The absolute ages for the Fourchu and Bourinot poles are after Van Eysinga (1975).

*These results are from three separate localities in the Marystown Group; they are denoted by a single symbol, MG (Fig. 9).

tions of these three areas suggested on the basis of geologic evidence (Schenk 1972; Rast and Skehan 1983). For the comparison between Avalon and North America, predicted paleolatitudes were calculated for a reference point corresponding to the present position of Cape Breton Island. In addition, we assumed in each subsequent comparison that Avalon remained fixed relative to that particular continent during the time interval compared. If some of our assumptions are incorrect, then this may be revealed as significant differences between observed and predicted paleolatitudes.

The paleolatitudes plotted versus age in Fig. 9, include both observed and predicted values for Avalon, as well as their associated latitude and age errors. The paleomagnetic data for each of the major continents are taken from recent compilations by other authors, but some newer data are added, as indicated in the caption.

In the comparison between Avalon and North America, we note that reliable data from North America for the interval 650–600 Ma are absent; on the other hand, numerous poles are available between 600 and 400 Ma, and these predict uniformly low paleolatitudes for the reference position of Avalon in the northern Appalachians. There is a considerable range in the predicted paleolatitude for any given time, particularly in the Cambrian, but the majority of data indicate paleolatitudes of less than 30°, and the maximum predicted values, including errors, are less than 35°. The range of paleolatitudes predicted for the reference locality clearly does not overlap with the observed intermediate paleolatitudes for Avalon. Therefore, we conclude that our assumption of adjacent positions of Avalon and North America, as in the Bullard *et al.* (1965) fit, is incorrect and that the two blocks occupied quite separate latitudes for the time interval 600–400 Ma. The difference between observed and predicted paleolatitudes for any time within this interval is, in fact, an estimate of the minimum separation of the reference locality and the position of Avalon.

The paleomagnetic data from the Armorican Massif are concentrated mainly in the range 640–480 Ma, and for this time they predict paleolatitudes for Avalon (in the Bullard *et al.* (1965) fit) that increase from near equatorial to ~60°S. A large variation in predicted paleolatitudes (0–60°) is apparent throughout this interval; however, this undoubtedly reflects, in part, the large age uncertainty and consequent overlap of ages for many of the pole data. Rather high paleolatitudes (~80°S) are predicted from a couple of Cambrian poles, whereas sparse but reliable data coverage for the interval 520–440 Ma indicates paleolatitudes that remained at about 60°S until the end of the Ordovician. As noted before, the paleolatitudes predicted from Armorica contrast sharply with the nearly constant equatorial paleolatitudes predicted from North American data (Van der Voo 1982; Peroud *et al.* 1984), whereas the observed paleolatitudes for Avalon and those predicted from Armorica reveal a fairly complete overlap for the interval ~620–450 Ma. This general agreement between Avalon and Armorica thus shows that their juxtaposition (as in the Bullard *et al.* (1965) fit) represents a paleomagnetically permissible reconstruction of these continents for the interval ~620–450 Ma.

Almost all of the paleomagnetic data available from Gondwana for this time period have ages between 600 and 480 Ma. Although the paleolatitudes predicted for Avalon from this data base again show a large variation (ranging from 0 to 75°S), a majority of the data is concentrated in a much smaller paleolatitude range, with predicted values for Avalon between 40 and 70°S. As was the case in the Avalon–Armorica comparison, the observed paleolatitudes overlap fairly well for the interval 600–480 Ma, suggesting similar positions, plausibly in juxtaposition, for Avalon and Gondwana. Even so, in view of the large error associated with the predicted paleolatitudes, it is clear that a Bullard *et al.* (1965) fit between Avalon and Gondwana is only one of a larger number of paleomagnetically permissible reconstructions.

Conclusions

Felsic volcanoclastic sediments of the Fourchu Group, and a gabbroic intrusion from the same region in southeastern Cape Breton Island, yield two often-superimposed, postfolding yet ancient components of magnetization, herein called A and B. The B component, with southerly declination and shallow inclination (mean D/I= 164°/16°) and lower blocking temperatures (<585°C), is commonly observed in the

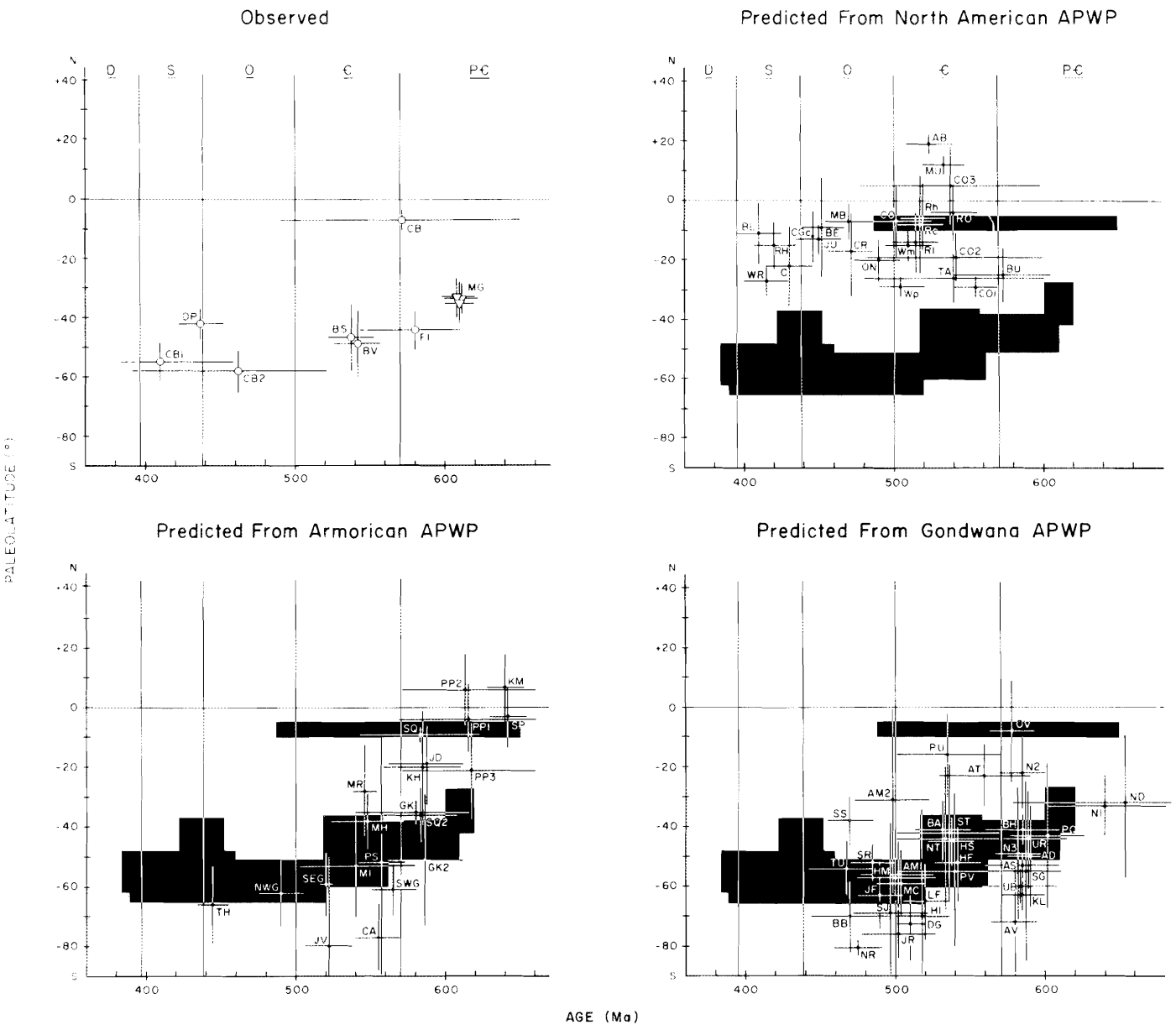


FIG. 9. Late Precambrian and early Paleozoic paleolatitudes plotted versus age for the Avalon terrane. The observed paleolatitudes for Avalon, from Table 3, are represented by open symbols in the upper left diagram, and their age and paleolatitude errors are shown by the horizontal and vertical error bars, respectively; note that the Bourinot poles have been assigned arbitrary age uncertainties of 30 Ma. Predicted paleolatitudes, represented by error crosses in the remaining diagrams, were calculated from available paleomagnetic data for the continents shown, assuming the fit of Bullard *et al.* (1965) between Avalon and each respective continent. The paleolatitude versus age field corresponding to the observed data for Avalon is superimposed on successive graphs of predicted paleolatitudes (dark-shaded region) for graphical comparison. Note that the geologic periods represented are labeled in the top two diagrams and that their absolute ages are after Van Eysinga (1975). The majority of the paleomagnetic data from other continents, including ages, was taken from lists compiled by other authors. For those poles whose age range was specified (e.g., stratigraphically dated poles), the error bars span the entire interval. The error for predicted paleolatitudes corresponds to A_{95} (McElhinny 1973), which in a few cases was calculated by us. The paleomagnetic data for North America are from Watts *et al.* (1980) (Cambrian only) and Van der Voo (1981) and also include the following recent additions: BU, Buckingham volcanics, all sites (Dankers and Lapointe 1981); ON, Oneota dolomite (Jackson and Van der Voo 1985); CGc, Cordova gabbro, C component (Dunlop and Stirling 1985); WR, Wabash reef (McCabe *et al.* 1985). The data for Armorica are from Perigo *et al.* (1983) and also include the following: CA, Carteret red beds (Perroud *et al.* 1982); TH, Thouars Massif (Perroud and Van der Voo 1985). The data for Gondwana have been listed by Hagstrum *et al.* (1980).

northern Appalachians and is probably a viscous partial thermoremanent magnetization (VpTRM) acquired during the late Paleozoic (Kiaman) interval, resulting from the tectonism during the Alleghenian orogeny. The A component has much steeper inclinations and dual polarity (mean $D/I = 132°/-63°$) and always has higher blocking temperatures ($>585°C$). This component, which obviously resides in hematite, is interpreted to have been acquired during oxidation processes associated with uplift and subaerial exposure of the Fourchu Group following the latest Precambrian Avalonian orogeny. Although paleomagnetic data for Avalon are still sparse, it is possible to perform paleomagnetic tests, using paleolatitudes only, on

proposed early Paleozoic reconstructions of Avalon and other Atlantic-bordering continents. The intermediate paleolatitudes corresponding to directions observed in Late Precambrian through Ordovician age rocks from Avalon show that it could not have been part of North America during this time. Although there is an inevitable uncertainty in the paleolatitude versus age data from the Armorican Massif and Gondwana, it appears that northwestern Africa and the Armorican Massif were at paleolatitudes similar to those occupied by Avalon. Thus, a fit of Avalon together with these areas for the interval from ~600 to 450 Ma is paleomagnetically permissible, corroborating reconstructions that have been suggested based on geologic and paleontologic evidence.

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